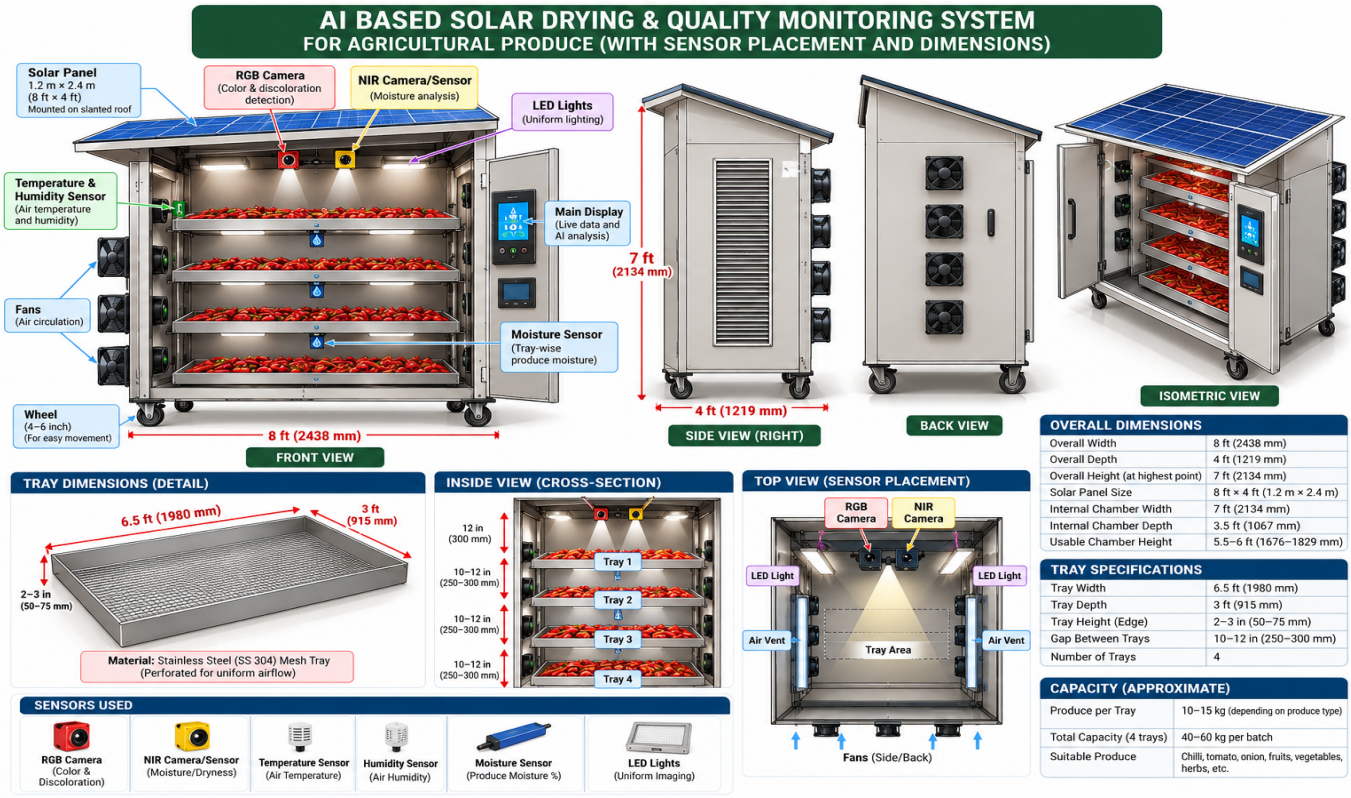
AI-BASED SOLAR DRYING & QUALITY MONITORING SYSTEM

For Agricultural Produce — Product Design, Measurements, Capacity, Sensors and AI Monitoring



OVERALL DIMENSIONS	
Overall Width	8 ft (2438 mm)
Overall Depth	4 ft (1219 mm)
Overall Height (at highest point)	7 ft (2134 mm)
Solar Panel Size	8 ft × 4 ft (1.2 m × 2.4 m)
Internal Chamber Width	7 ft (2134 mm)
Internal Chamber Depth	3.5 ft (1067 mm)
Usable Chamber Height	5.5–6 ft (1676–1829 mm)

TRAY SPECIFICATIONS	
Tray Width	6.5 ft (1980 mm)
Tray Depth	3 ft (915 mm)
Tray Height (Edge)	2–3 in (50–75 mm)
Gap Between Trays	10–12 in (250–300 mm)
Number of Trays	4

CAPACITY (APPROXIMATE)	
Produce per Tray	10–15 kg (depending on produce type)
Total Capacity (4 trays)	40–60 kg per batch
Suitable Produce	Chilli, tomato, onion, fruits, vegetables, herbs, etc.

Technical product concept and proposed prototype specification

Purpose: A solar-powered drying/storage system enhanced with RGB and NIR imaging, moisture measurement, temperature/humidity sensing, controlled lighting, airflow and AI-based analysis to estimate produce dryness/moisture and detect discoloration.

Source drawing basis: The supplied original drawing identifies an 8 ft front width and 4 ft side depth and shows trays, heat/moisture sensing, displays, fans, air vents, wheels and a solar panel. The 7 ft overall height, tray dimensions and 40–60 kg capacity in this report are **proposed prototype design values**, not dimensions stated in the original drawing.

1. Product Overview

The proposed system combines the existing solar drying/storage concept with an AI inspection layer. Agricultural produce is placed on multiple perforated trays. Fans circulate air through the chamber while sensors record environmental and moisture information. A top-mounted RGB camera captures visible colour and surface changes, while an NIR camera/sensor is proposed to support moisture/dryness analysis. Controlled LED illumination keeps image conditions consistent.

The AI/software layer receives the sensor and image data and can produce moisture content estimates, dryness level, discoloration percentage and an overall quality/grade indicator. A display or web/mobile dashboard can present the results.

2. Overall Product Dimensions

Parameter	Proposed specification	Basis / note
Overall width	8 ft (2438 mm)	Shown in supplied original front-view drawing
Overall depth	4 ft (1219 mm)	Shown in supplied original side-view drawing
Overall height	7 ft (2134 mm)	Proposed prototype value; not specified in source drawing
Internal chamber width	Approx. 7 ft (2134 mm)	Proposed usable internal dimension
Internal chamber depth	Approx. 3.5 ft (1067 mm)	Proposed usable internal dimension
Usable chamber height	Approx. 5.5–6 ft (1676–1829 mm)	Proposed; depends on frame/roof construction
Number of trays	4	Proposed layout based on supplied multi-tray concept
Wheel diameter	Approx. 4–6 in	Proposed heavy-duty mobility specification

Important: The supplied drawing explicitly shows 8 ft and 4 ft, but does not provide an overall height, exact tray size, tray spacing or tested capacity. Those values are therefore presented here as engineering targets for a prototype and should be validated during fabrication.

3. Tray Design and Capacity

A four-tray configuration is recommended for the prototype. The trays should be perforated/mesh to allow airflow from the fans and should use food-grade stainless steel (for example SS 304) where practical. The load should be spread in a controlled layer rather than piled deeply so that the camera can observe the surface and air can pass through the produce.

Tray parameter	Proposed measurement
Tray width	6.5 ft (1980 mm)
Tray depth	3 ft (915 mm)
Tray height / edge	2–3 in (50–75 mm)
Gap between trays	10–12 in (250–300 mm)
Number of trays	4
Recommended material	Food-grade SS 304 perforated/mesh

Capacity item	Target / estimate
Produce per tray	10–15 kg, depending on produce and loading thickness
Total for 4 trays	Approx. 40–60 kg fresh produce per batch
Capacity status	Prototype design target; requires physical validation
Suitable examples	Chilli, tomato, onion, fruits and vegetables

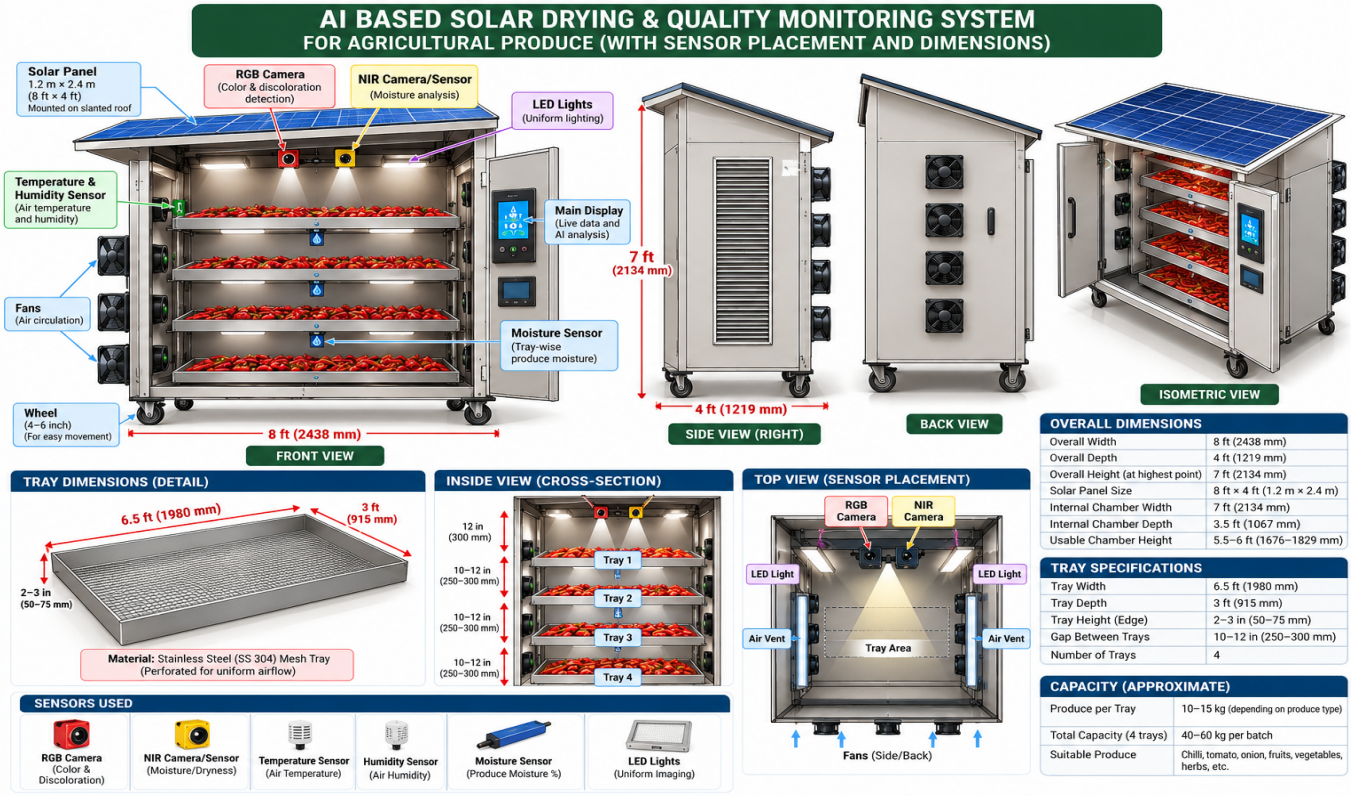


Figure: Proposed full product dimensions, tray dimensions, sensor placement and capacity.

4. Sensor and Camera Configuration

Component	Placement	Purpose
RGB camera	Top/ceiling, centered over tray area	Visible colour, texture and discoloration detection
NIR camera/sensor	Beside RGB camera	Moisture/dryness-related spectral information
LED lights	Around camera field of view	Uniform, repeatable illumination for imaging
Temperature sensor	Side wall, away from direct fan outlet	Air/chamber temperature
Humidity sensor	Near temperature sensor	Relative humidity during drying
Produce moisture sensor/meter	Tray-level / removable measurement point	Actual moisture % for calibration/ground truth
Fans	Side/back walls	Air circulation and uniform drying
Solar panel + battery	Roof / upper structure	System power

Recommended imaging arrangement: RGB + NIR at the top, looking downward; LED lighting around the cameras; temperature/humidity on the chamber side wall; moisture measurement at tray level. The camera should not be mounted directly next to a hot-air outlet or vibrating fan.



Figure: Product concept showing camera, NIR, moisture, environmental sensors and AI output.

5. Hardware, Software and AI Architecture

Hardware layer: solar panel → battery/power management → controller (ESP32/Raspberry Pi or similar) → fans, sensors, cameras and display. The original concept already includes solar power, fan operation, sensor reading and moisture control.

Data layer: every observation should store an image, timestamp, tray/sample identifier, temperature, humidity and measured moisture percentage. This creates a dataset that can be used to train and validate the AI model.

AI layer: use computer vision to identify discoloration regions and regression/classification methods to estimate dryness/moisture and quality grade. NIR data can be added later to improve moisture-related prediction.

Software output: moisture %, dryness level, discoloration %, quality grade, drying progress and alerts. Results can be shown on a local display and/or web/mobile dashboard.

Step	Process
1	Load agricultural produce onto trays
2	Start solar-powered drying and airflow
3	Capture RGB + NIR images under controlled LED lighting
4	Record temperature, humidity and measured moisture
5	AI analyses images and sensor data
6	Estimate moisture/dryness and quantify discoloration
7	Display result and store data for model improvement

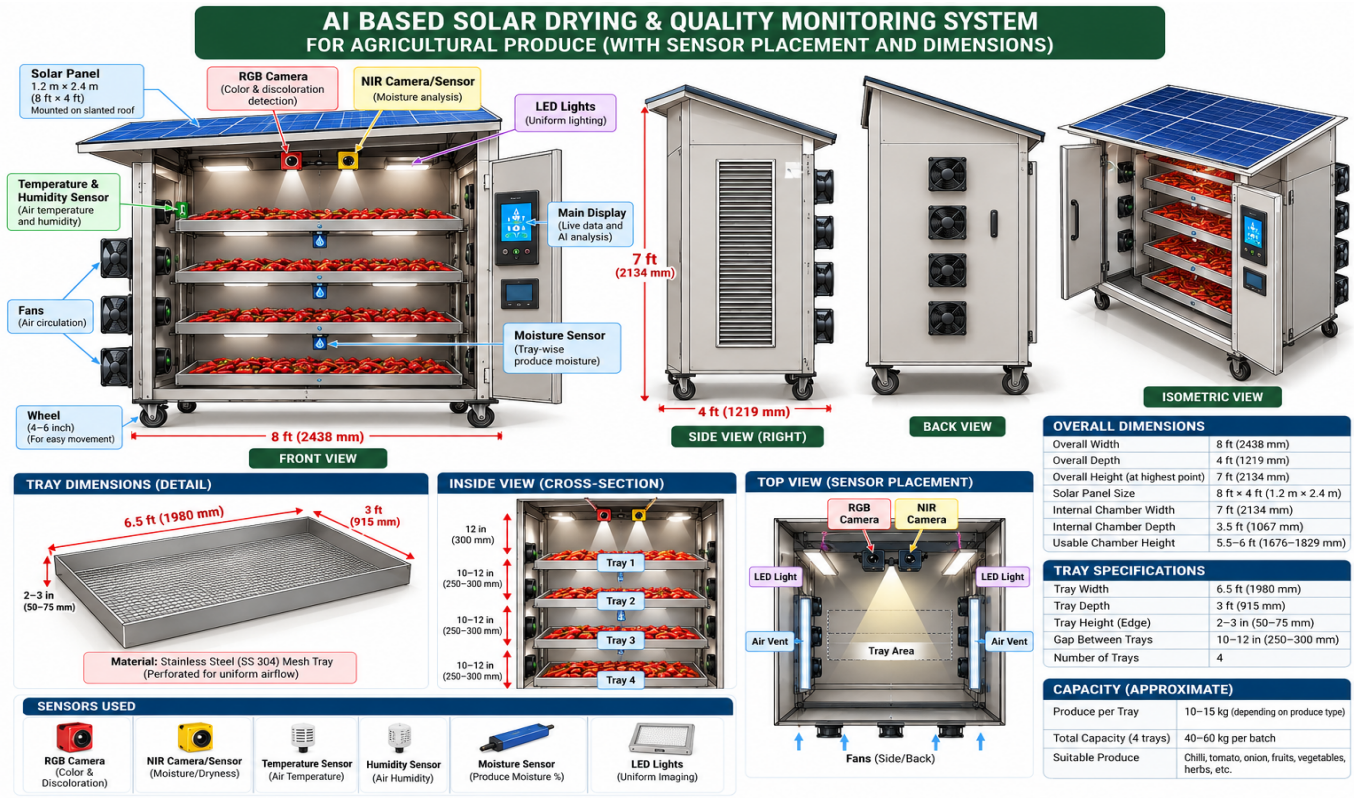


Figure: Integrated product layout and system architecture.

6. Original Product Drawings Supplied for the Project

The following pages reproduce all four figures from the supplied drawing. They are retained as the source design basis. Fig. 1 shows the front and side views with the 8 ft width and 4 ft depth; Fig. 2 shows the back/side arrangement, fans and tray wheels; Fig. 3 shows the solar panel, automatic moisture regulator, rotating switch and fans; Fig. 4 shows the power/control and moisture decision workflow.

A LOW-COST AFFORDABLE CUSTOMIZED STORAGE SYSTEM FOR IMPROVING SHELF LIFE OF AGRICULTURAL PRODUCE

DRAWINGS:

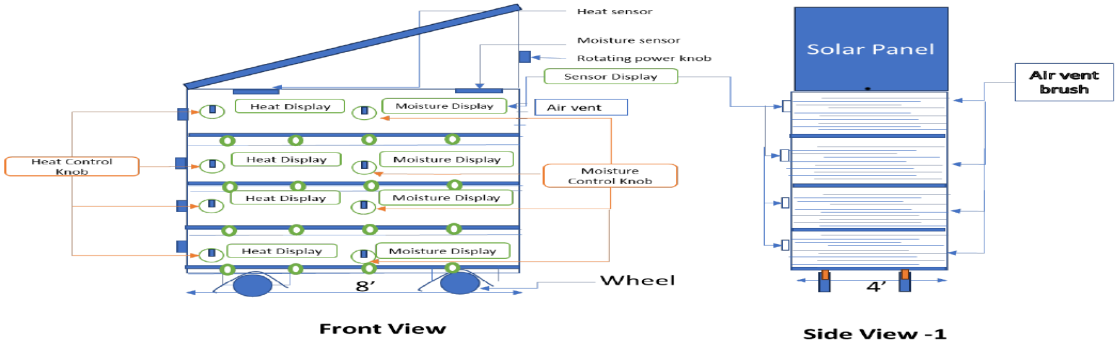


FIG. 1

Original Drawing — Figure 1

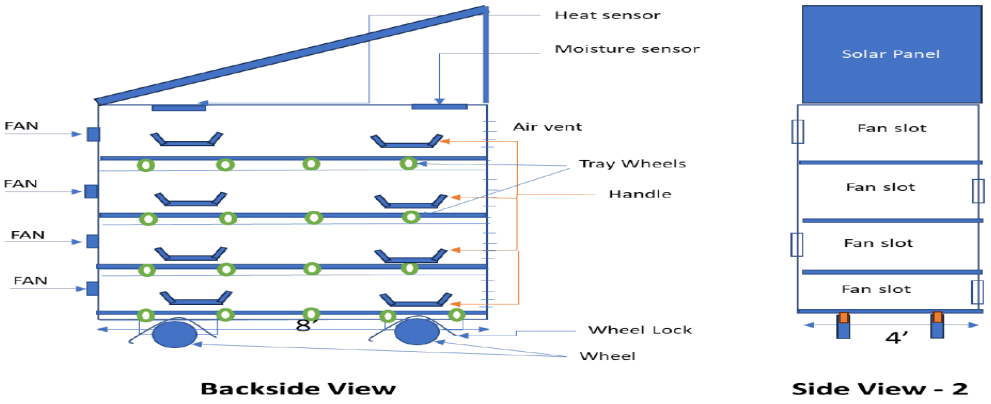


FIG. 2

Original Drawing — Figure 2

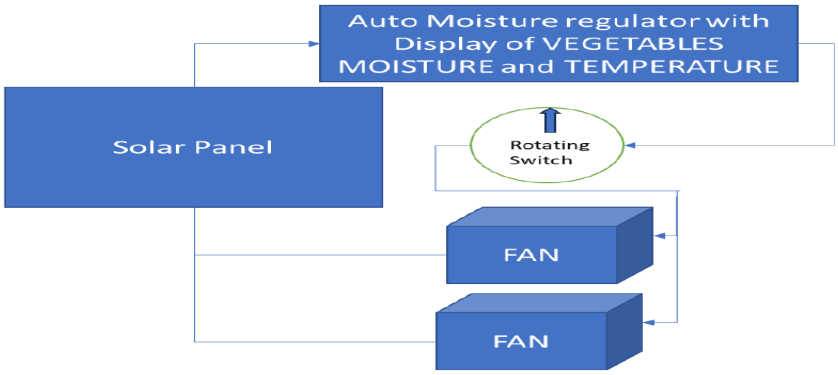


FIG. 3

Original Drawing — Figure 3

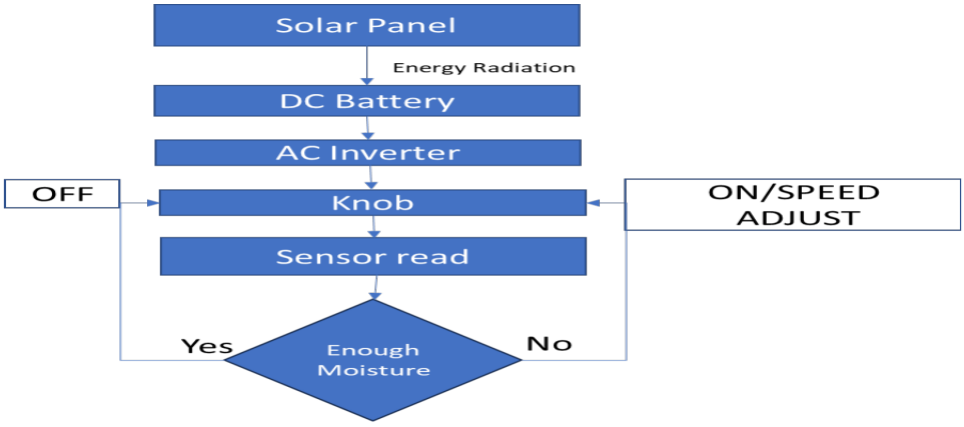


FIG. 4

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Original Drawing — Figure 4

7. Consolidated Product Specification

Category	Specification
Product type	Solar-powered agricultural drying + AI quality monitoring system
Overall size	8 ft W × 4 ft D × 7 ft H (proposed height)
Drying trays	4 trays
Tray size	6.5 ft W × 3 ft D × 2–3 in edge
Tray spacing	10–12 in
Target capacity	40–60 kg fresh produce per batch
Imaging	RGB camera + NIR camera/sensor
Lighting	Controlled LED illumination
Environment sensing	Temperature + humidity
Moisture measurement	Produce moisture sensor/meter for ground truth/calibration
Airflow	Multiple fans and vents
Power	Solar panel + battery; controller and electrical power management
AI outputs	Moisture estimate, dryness level, discoloration %, quality grade
Software	Local display plus optional web/mobile dashboard
Mobility	Four wheels with wheel locks

Design note: The original source drawing supports the 8 ft width, 4 ft depth and the listed functional elements. The additional 7 ft height, tray dimensions and 40–60 kg capacity are proposed prototype targets for development. Before fabrication, these values should be checked against structural loads, tray material, airflow, solar power budget, camera field of view and the selected agricultural produce.

End of Report